



Web Development Foundations

Week 3 - "Layout: Control Space"
Day 2: "Responsive Iteration"

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Reconnect

Monday:

- ▶ every element occupies space
- ▶ containers can control children
- ▶ Flexbox creates a first layout

Today:

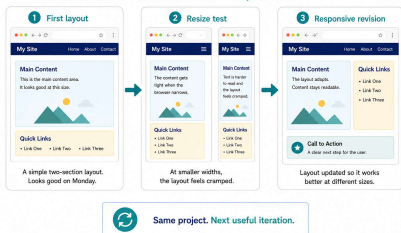
- ▶ test the layout under pressure
- ▶ revise it for a narrower screen

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From Monday to Wednesday

Iterate, test, and improve.



Same project. Next useful iteration.

Reconnect

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The Working Problem: One Size Is Not Enough

A layout can work at one size and fail at another.

Common signs:

- ▶ content feels squeezed
- ▶ navigation wraps awkwardly
- ▶ cards become too narrow
- ▶ text becomes hard to scan
- ▶ the user has to fight the page

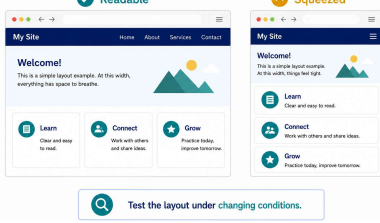


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One Layout Size Is Not Enough

The same page can feel different at different widths.



Test the layout under changing conditions.

One Size - Not Enough

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What Counts as Success Today

- you test the page at more than one width
- you notice one layout problem
- one media query changes the layout
- content remains readable
- you can explain why the change helps

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Responsive Does Not Mean Smaller

Responsive design means the layout adapts.

It may:

- ▶ stack columns
- ▶ change spacing
- ▶ simplify navigation
- ▶ adjust card width
- ▶ preserve readability

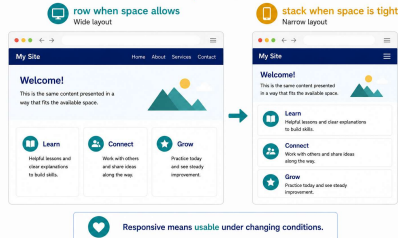
The goal is usability under changing conditions.



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Responsive Design Is Adaptation, Not Shrinking

The content stays readable and usable.



Responsive = Adapt

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What We Will Use Today

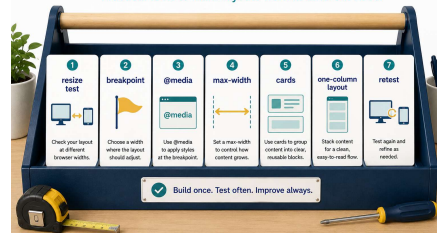
- ▶ resize test
- ▶ breakpoint
- ▶ '@media'
- ▶ 'max-width'
- ▶ card layout
- ▶ one-column layout
- ▶ inspect, revise, retest



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Today's Responsive-Layout Toolbox

Practical tools to make layouts work at different sizes.



Today's Toolbox

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Breakpoints Are Conditions

A breakpoint says:

"When the screen meets this condition, use these CSS rules."

Example:

- ▶ wide screen: cards can sit in a row
- ▶ narrow screen: cards may need to stack

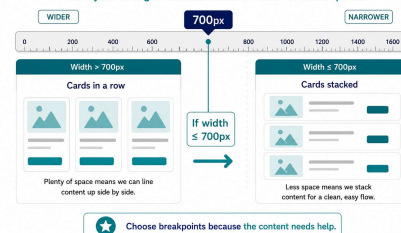
Choose breakpoints because the content needs help.



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Breakpoint = A Condition

The layout changes when the width reaches a certain point.



Choose breakpoints because the content needs help.

Breakpoint As Condition

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Media Queries, Gently

Small pattern:

```
# Code_Samples.com
1 @media (max-width: 700px) {
2   .card-grid {
3     grid-template-columns: 1fr;
4   }
5 }
6
```

Read it as:
"At 700px or smaller, stack the cards."

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Demo

"Responsive Cards"

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What to Watch For

- ▶ desktop layout first
- ▶ resize before the fix
- ▶ name the usability problem
- ▶ add the media query
- ▶ resize again
- ▶ confirm the HTML stayed the same

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Demo: Responsive Cards

Same content. Better layout at different widths.

Wide Width (before media query) → Narrow Width (after media query)

Student Resources

Library: Find books, articles, and research help. [Learn More](#)

Tutoring: Get one-on-one help and study support. [Learn More](#)

Technology: Borrow equipment and find tech help. [Learn More](#)

Add a media query: @media

Resize, revise, retest.

Demo: Responsive Cards

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Inspect, Revise, Retest

Responsive work is a loop:

1. inspect the current layout
2. resize the browser
3. identify one problem
4. revise one rule
5. retest the page

Do not fix five problems at once!

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Responsive Design Workflow

A simple loop for better layouts at every size.

Do not fix five problems at once.
Small changes. Clear results. Better layouts.

Responsive Design Workflow

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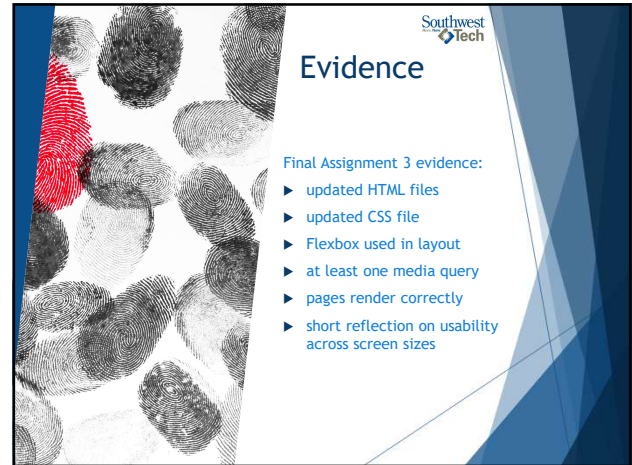


Lab Refinement

Responsive refinement
Your goal:

- ▶ test your site at different widths
- ▶ add at least one media query
- ▶ improve spacing or alignment
- ▶ keep navigation usable
- ▶ make the layout adapt, not just shrink

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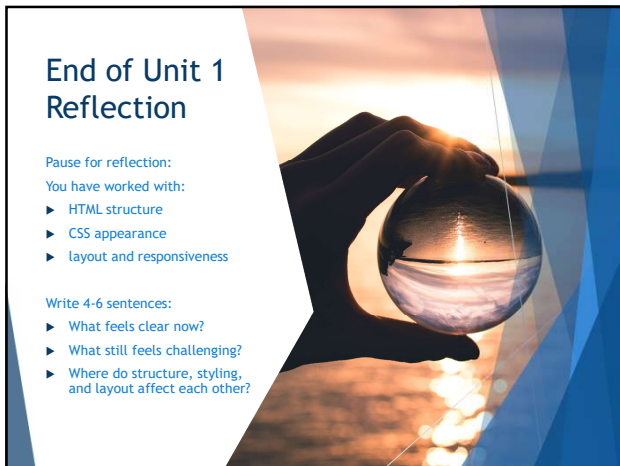


Evidence

Final Assignment 3 evidence:

- ▶ updated HTML files
- ▶ updated CSS file
- ▶ Flexbox used in layout
- ▶ at least one media query
- ▶ pages render correctly
- ▶ short reflection on usability across screen sizes

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End of Unit 1 Reflection

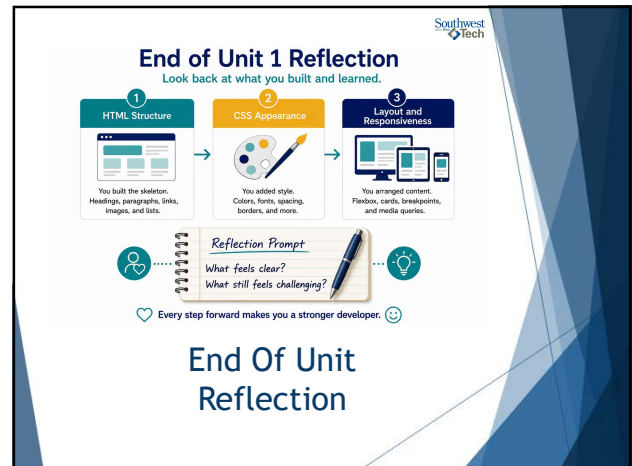
Pause for reflection:
You have worked with:

- ▶ HTML structure
- ▶ CSS appearance
- ▶ layout and responsiveness

Write 4-6 sentences:

- ▶ What feels clear now?
- ▶ What still feels challenging?
- ▶ Where do structure, styling, and layout affect each other?

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End of Unit 1 Reflection
Look back at what you built and learned.

1 **HTML Structure**
You built the skeleton. Headings, paragraphs, links, images, and lists.

2 **CSS Appearance**
You added style. Colors, fonts, spacing, borders, and more.

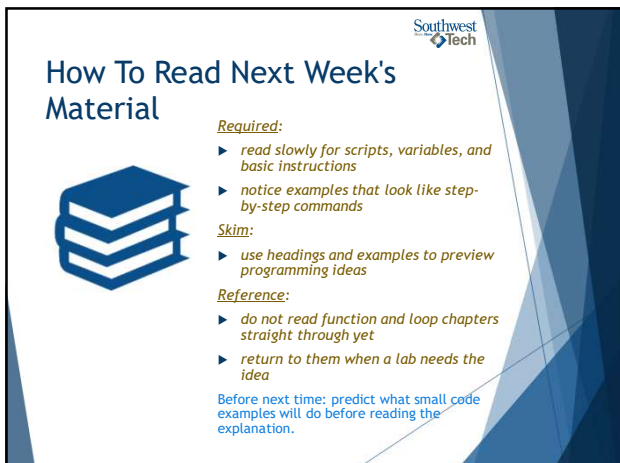
3 **Layout and Responsiveness**
You arranged content. Flexbox, cards, focalpoints, and media queries.

Reflection Prompt
What feels clear?
What still feels challenging?

Every step forward makes you a stronger developer.

End Of Unit Reflection

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How To Read Next Week's Material

Required:

- ▶ read slowly for scripts, variables, and basic instructions
- ▶ notice examples that look like step-by-step commands

Skim:

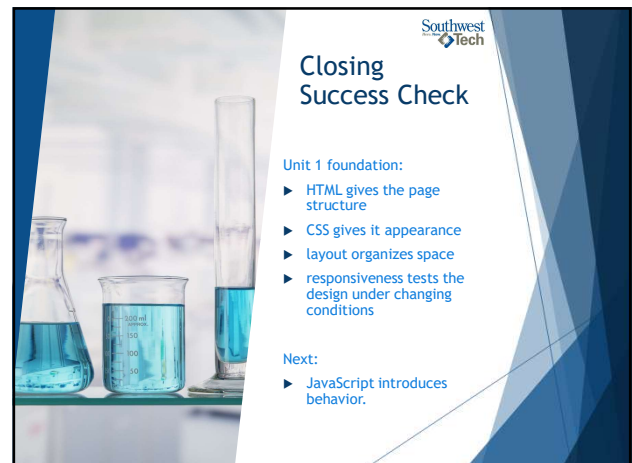
- ▶ use headings and examples to preview programming ideas

Reference:

- ▶ do not read function and loop chapters straight through yet
- ▶ return to them when a lab needs the idea

Before next time: predict what small code examples will do before reading the explanation.

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Closing Success Check

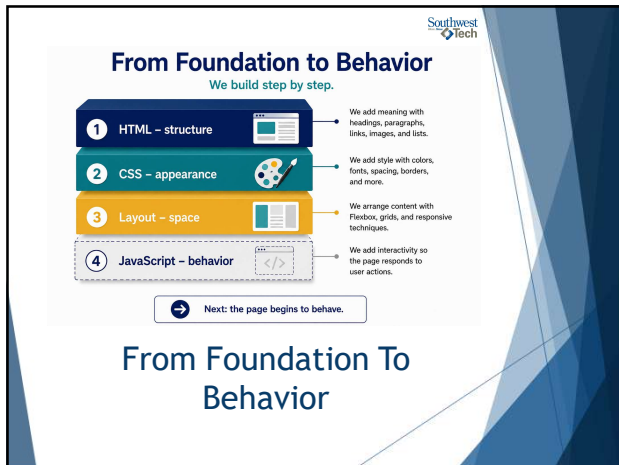
Unit 1 foundation:

- ▶ HTML gives the page structure
- ▶ CSS gives it appearance
- ▶ layout organizes space
- ▶ responsiveness tests the design under changing conditions

Next:

- ▶ JavaScript introduces behavior.

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